

MT 501 : Energy Conversion & Storage Technology

Course Code : MT 501

Course Name : Energy Conversion & Storage Technology

L-P-T-C: 3-0-0-3

Intended for: B.Tech./M.Tech./M.Sc./Ph.D

Prerequisites: None

Mutual Exclusion: EN 503

Approval : 57th BoA

Course Contents

- **Energy conversion & storage systems overview:** Scope of energy storage & conversion, needs and opportunities in energy storage, Overview of key technologies and interdisciplinary aspects, with a focus on comparing the time scales of various storage methods and their application, Energy storage in the power and transportation sectors. Importance of energy conversion systems in the development and performance of electric vehicles. (5 Hours)
- **Energy Conversion Systems:** Energy conversion systems transform various forms of energy—such as mechanical, thermal, chemical, or solar—into electrical energy. (6 Hours)
- **Chemical Energy Storage Systems:** Chemical storage system- hydrogen, methane, etc., concept of chemical storage, advantages and limitations, challenges, and future prospects of chemical storage systems. (5 Hours)
- **Thermal Storage Systems:** thermal storage systems such as heat pumps, hot water storage tanks, and solar thermal collectors. Examine the application of phase change materials (organic and inorganic) for heat storage, along with system efficiencies and economic assessments. (6 Hours)
- **Electromagnetic Energy Storage Systems:** Electromagnetic storage systems, including double-layer capacitors for electrostatic charge storage and superconducting magnetic energy storage (SMES). Analyze the concepts, advantages, limitations, and future prospects of these systems. (5 Hours)
- **Electrochemical storage and conversion systems:**
 - Batteries-Working principle of battery, primary and secondary (flow) batteries, battery performance evaluation methods, major battery chemistries and their voltages- Li-ion battery & Metal hydride battery vs lead-acid battery, metal-air battery. (6 Hours)
 - Supercapacitors- Working principle of supercapacitor, types of supercapacitors, cycling and performance characteristics, difference between battery and supercapacitors, Introduction to Hybrid electrochemical supercapacitors (6 Hours)
 - Fuel cell: Operational principle of a fuel cell, types of fuel cells, hybrid fuel cell-battery systems, hybrid fuel cell-supercapacitor systems. (3 Hours)
 - Water electrolysis for hydrogen generation: Cell technologies, Gibbs Free energy of the water splitting reaction and cell voltage, Current-Voltage relationships, Degradation mechanisms. (3 Hours)
- **Battery design for transportation:** Mechanical design and packaging of battery packs for electric vehicles, advanced quick charging methods, thermal management, state of charge and state of health estimation, and the recycling of batteries used in electric vehicles. (4 Hours)



Laboratory:

1. NA

Text books:

1. Frank S. Barnes and Jonah G. Levine, **Large Energy Storage Systems Handbook**, CRC press, 2011.
2. Ralph Zito, **Energy storage: A new approach**, Wiley, 2010.

References:

1. Pistoia, Gianfranco, and Boryann Liaw, **Behaviour of Lithium-Ion Batteries in Electric Vehicles: Battery Health, Performance, Safety, and Cost**, Springer International Publishing AG, 2018.
2. Robert A. Huggins, **Energy storage**, Springer Science & Business Media, 2010.